

PALGUN THATIPAMULA

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[LinkedIn](#) — [GitHub](#) — [Google Scholar](#)

PROFESSIONAL SUMMARY

M.Tech AI researcher at DIAT-DRDO specializing in Computer Vision, Edge AI, and Counter-UAS systems. Experienced in designing real-time multi-object tracking and anomaly detection pipelines optimized for resource-constrained tactical hardware. Lead author of an IEEE conference publication and patent co-inventor with expertise in YOLO-based perception systems, spatiotemporal learning, and deployable deep learning architectures.

EDUCATION

Defence Institute of Advanced Technology (DIAT-DRDO)

M.Tech in Artificial Intelligence (CSE) — CGPA: 8.19/10

Rajiv Gandhi University of Knowledge Technologies (RGUKT)

B.Tech in Electronics & Communication Engineering — CGPA: 7.85/10

Pune, Maharashtra

2024 – Present

Basar, Telangana

2020 – 2024

EXPERIENCE

Student Intern — MeitY Sponsored Research Project

Defence Institute of Advanced Technology (DIAT-DRDO)

Pune, Maharashtra

Mar 2026 – Present

- Contributing to the “ExxS” Explainable AI (XAI) research initiative focused on interpretable deep learning systems.
- Exploring feature attribution and interpretability techniques for improving trustworthiness in mission-critical AI applications.

KEY PROJECTS

Drone Anomaly Detection Pipeline

YOLOv8n, ConvLSTM — 48GB NVIDIA RTX A6000

- Developed a real-time drone tracking and anomaly detection pipeline capable of distinguishing UAV motion from biological noise.
- Achieved 0.944 AUROC using a cascaded YOLOv8n + ConvLSTM architecture trained on spatiotemporal flight behavior.
- Optimized inference throughput for real-time tactical surveillance environments.

Edge-AI UAV Kinematic Surveillance

NVIDIA Jetson Orin — Spatiotemporal Fusion

- Designed a shared-latent multi-task tracking architecture for UAV intent recognition on resource-constrained edge hardware.
- Achieved 94.8% intent classification accuracy while maintaining VRAM utilization strictly below 150MB.
- Demonstrated the system during Anti-UAV technical sessions conducted for CISF officers (April 2026).

Physics-Based Drone & Marine Engine Digital Twins

Deep Learning, Sensor Fusion

- Developed a physics-based digital twin for UAVs to simulate real-time flight dynamics and monitor structural parameters.
- Built a predictive maintenance framework for marine engines using sensor data for continuous operational state monitoring.

PUBLICATIONS & INTELLECTUAL PROPERTY

Publications:

- Palgun Thatipamula (**Main Author**), “Real-Time Cascaded Pipeline for Drone Detection and Behavior Anomaly Analysis: YOLOv8 Benchmark and ConvLSTM Approach,” IEEE PuneCon 2025.

Intellectual Property:

- Co-inventor**, Computer vision-based player tracking and performance analysis system patent (Granted May 2026).

TECHNICAL SKILLS

AI/ML: Computer Vision, Edge AI, Explainable AI (XAI), Multi-Object Tracking, Spatiotemporal Fusion, Deep Learning, Object Detection.

Frameworks & Models: PyTorch, OpenCV, TensorRT, ONNX, YOLOv8n, ConvLSTM, ViT, GRU, LSTM.

Hardware & Deployment: NVIDIA Jetson Orin, NVIDIA RTX A6000, Model Quantization, Real-Time Inference, Low-SWaP Optimization, Edge Deployment, CUDA, Linux.

Programming: Python, C, Java, MATLAB, Verilog.

Soft Skills: Technical Mentorship, Research Collaboration.

ACHIEVEMENTS & HONORS

- Competitive Exams:** Qualified UGC-NET 2025 (CSE), GATE 2024 (AIR 5893 in ECE; also qualified CSE), and TG PGECET 2024 (Rank 142 in ECE).
- Exhibitions & Speaking:** Demonstrated drone research at GMRT National Science Day (2026); presented at IConIC 2024 (Optic Disc Detection); conducted a CV workshop as a Session Speaker at Panimalar Engineering College.
- Workshops:** Participated in SENSE 2025 (Smart Electronics & Next-Gen Systems).
- Athletics:** National-level Table Tennis player and 5-time university champion.

RESEARCH INTERESTS

Computer Vision, Edge AI, Counter-UAS Systems, Explainable AI, Spatiotemporal Learning, Autonomous Surveillance Systems.